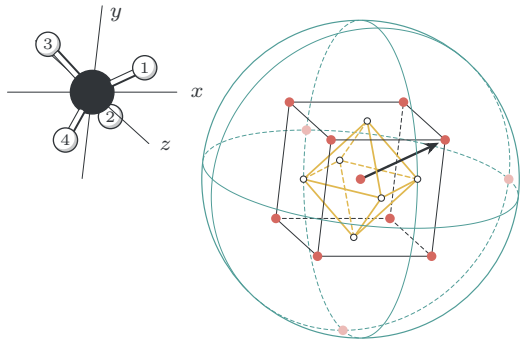
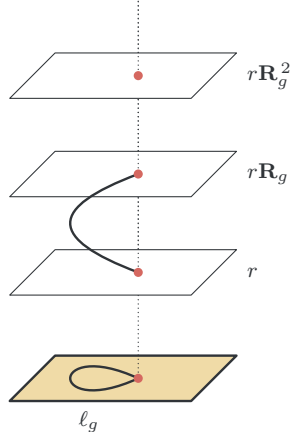


Methane,  $\text{CH}_4$ : the reference  $\mathbf{X}_0$  in its body frame,  
with the  $C_2$  axes  $x, y, z$  through the midpoints of opposite edges.



The orientation space  $SO(3)$  as a ball:  
direction the axis, distance the angle.  
Red dots are the twelve rotations of  $T$ ;  
the gold octahedron is the cell  $\mathcal{U}$  at  $q = 0$ ;  
the arrow is the lift  $r \rightarrow r\mathbf{R}_g$  of the  
third-turn about bond 1.



Over the base  $SO(3)/T$  lie twelve sheets,  
one for each rotation in  $T$ ; three are drawn.  
The third-turn is a closed loop  $\ell_g$  in the base;  
lifted to the sheets it does not close,  
but climbs from sheet  $r$  to sheet  $r\mathbf{R}_g$ .  
A species sees the climb as a number or matrix:  
A as 1,  ${}^1\text{E}$  as  $\omega$ ,  ${}^2\text{E}$  as  $\omega^2$ , T as  $x \rightarrow y \rightarrow z$ .